

ROLLUP

★★★★★ Advanced Reporting Feature

ROLLUP is used when you want SQL to generate subtotals and a grand total automatically, without writing multiple queries.

Think of reports like:

- Sales by City → State → Country
- Revenue by Product → Category
- Employees by Team → Department
- Orders by Month → Year

Instead of manually calculating totals, ROLLUP does it for you.

14.17.1 Why ROLLUP Exists

Suppose we have a sales table.

Country	State	City	Sales
India	Delhi	New Delhi	100
India	Delhi	Dwarka	200
India	Bihar	Arrah	150
India	Bihar	Patna	250

Normal GROUP BY

```
SELECT state,
       city,
       SUM(sales)
FROM Sales
GROUP BY state, city;
```

Output

State	City	Sales
Delhi	New Delhi	100
Delhi	Dwarka	200
Bihar	Arrah	150
Bihar	Patna	250

Useful.

But business reports usually also need

- Delhi Total
- Bihar Total
- India Total

Without ROLLUP, we'd need several extra queries.

14.17.2 Syntax

```
SELECT columns,  
       aggregate  
FROM table  
GROUP BY ROLLUP(column1, column2, ...);
```

14.17.3 Two-Level ROLLUP

```
SELECT state,  
       city,  
       SUM(sales)  
FROM Sales  
GROUP BY ROLLUP(state, city);
```

Output

State	City	Sales
Bihar	Arrah	150
Bihar	Patna	250
Bihar	NULL	400
Delhi	Dwarka	200
Delhi	New Delhi	100
Delhi	NULL	300
NULL	NULL	700

Notice the NULL rows.
They are generated by SQL.

14.17.4 Understanding NULL Rows

```

Bihar
  Arrah      150
  Patna      250
-----
Bihar Total  400

Delhi
  Dwarka      200
  New Delhi   100
-----
Delhi Total  300

-----
Grand Total  700

```

The generated rows appear as

`City = NULL`

because there is no specific city.

ROLLUP follows hierarchy from left to right.

`ROLLUP(state, city)` means

State



City

SQL computes

State + City



State Total



Grand Total

What is ROLLUP?

Generates hierarchical subtotals and a grand total.

Why does ROLLUP produce NULL rows?

NULL indicates a subtotal or total level where that grouping column no longer has a single value.

Is ROLLUP hierarchical?

Yes.

Left-to-right hierarchy.

Does column order matter?

Absolutely.

`ROLLUP(state,city)` is different from `ROLLUP(city,state)`

Why CUBE Exists

Suppose

Department	City	Salary
ECE	Delhi	70
ECE	Mumbai	60
CSE	Delhi	90
CSE	Mumbai	80

Business may ask

- Salary by department and city
- Department totals
- City totals
- Overall total

Without CUBE

Need many separate queries.

14.18.2 Syntax

```
SELECT columns,
       aggregate
FROM table
GROUP BY CUBE(column1,column2);
```

14.18.3 GROUP BY Only

```
GROUP BY department,city
```

Produces

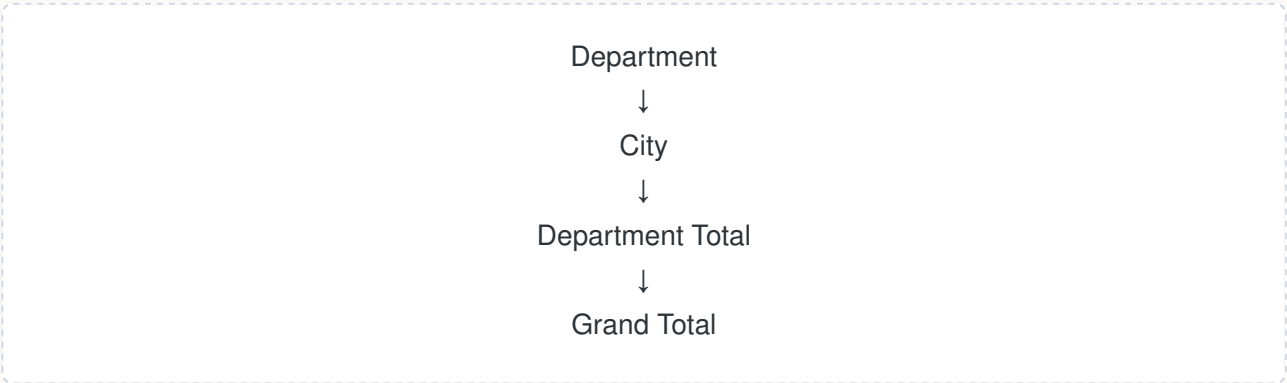
Department	City
ECE	Delhi
ECE	Mumbai
CSE	Delhi
CSE	Mumbai

Only detailed groups.

14.18.4 ROLLUP

```
ROLLUP(department,city)
```

Produces



Output

Department	City
ECE	Delhi
ECE	Mumbai
ECE	NULL
CSE	Delhi
CSE	Mumbai
CSE	NULL
NULL	NULL

Notice
No city totals.

14.18.5 CUBE

```
GROUP BY CUBE(department,city);
```

Output

Department	City
ECE	Delhi
ECE	Mumbai
CSE	Delhi
CSE	Mumbai
ECE	NULL
CSE	NULL
NULL	NULL
NULL	Delhi
NULL	Mumbai

Notice the extra rows

NULL
Delhi

These are
City Totals.

ROLLUP

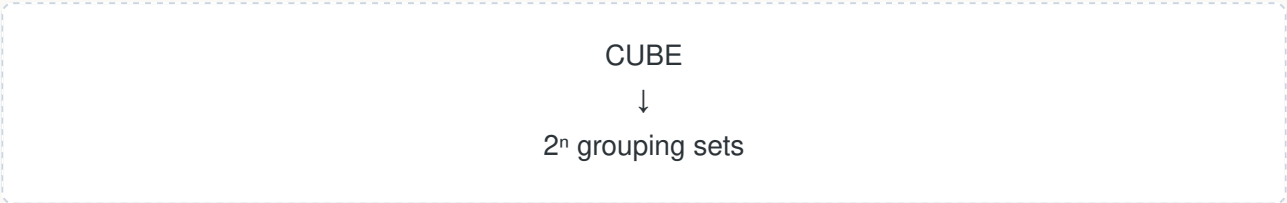
Department + City
↓
Department Total
↓
Grand Total

CUBE

Department + City
↓
Department Total
↓
City Total
↓
Grand Total

Number of Grouping Sets

For n columns:



Examples

Columns	Grouping Sets
1	2
2	4
3	8
4	16

This exponential growth is why large CUBEs can become expensive.

14.18.9 Performance

CUBE generates far more rows than ROLLUP.

Example
2 columns



With more columns, the difference grows rapidly.